

CP317A – Group Project

Project Title: Car Rental System

Deliverable: Sprint 3

Group ID: Team # 28

Team Members and Roles:

- SabilUllah Hussaini (Product Owner)
- Sahil Sandhu (Product Owner & Developer)
- Moosa Ahmed (Scrum Master)
- Mahad Farrukh (Developer)
- Ayesha Raheel (Developer)
- Kritika Kamath (Developer)
- Shalin Panjwani (Developer)

Sprint Goal

The goal of Sprint 3 is to bring together all of the major features created in the earlier sprints and make sure they work smoothly as a single system. This sprint focuses on connecting customer features, admin tools, database logic, and the user interface so that the application feels complete and stable. The work in this sprint prepares the project for the final submission and ensures that the system behaves correctly across different flows.

Sprint 1 gave us the authentication system and the search foundation, while Sprint 2 delivered the reservation workflow, shopping cart, filtering features and the dashboard. Sprint 3 builds on this by implementing the Admin Portal, completing the remaining integration work, improving data consistency and resolving bugs that appeared when the modules were connected together. This sprint also includes the early preparation for final documentation, testing and polishing the user experience.

Since ADMIN 1 and ADMIN 2 were moved from Sprint 2, this sprint focuses heavily on administrative features. By the end of Sprint 3, the system is expected to function as a full working prototype with both customer and admin sides connected.

Sprint Backlog

Sprint 3 contains five user stories that are focused on admin features, payment integration and full system refinement. The stories chosen match the priorities in Milestone 2 and the adjustments made at the end of Sprint 2.

Story ID	Story Title	Story Points	Priority	Assigned Team Members	Status	Key Tasks
ADMIN 1	Admin Login Portal	3	High	Mahad, Moosa, Sabil	Done	Create admin login interface, implement role validation, connect login to database, add credential checks, confirm admin access flow
ADMIN 2	Admin Vehicle Inventory Management	8	High	Ayesha, Kritika, Sahil	Done	Create CRUD interface, add vehicle add and update forms, implement delete and deactivate options, handle images, sync with user search results
UI 4 (Integration)	Connect search filters with full results flow	5	Medium	Sahil, Shalin	Done	Link filters to updated backend queries, connect filter selections to vehicle details pages, improve load time, check UI responsiveness
PAY 1	Simulated Payment Gateway Integration	5	High	Mahad, Sahil	Done	Connect payment form with booking logic, improve card validation, add payment error handling, update confirmation screen

REP 1	Admin Dashboard and KPIs	8	Medium	Moosa, Sabil	Done	Design KPI layout, implement booking statistics, calculate fleet usage, connect dashboard to admin portal
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Implementation and progress

During Sprint 3, the team focused mainly on integrating the different features created in the earlier sprints and completing the admin portal. The work completed in this sprint helped the system feel more complete and allowed us to move closer to a final working version of the project.

Code Snippets:

The screenshot displays the 'Car Rental Admin Panel' interface. At the top, there are navigation links: 'Edit Users', 'Edit Cars', 'View KPIs', 'Name Here', 'Exit to Dashboard', and 'Logout'. The main content area is titled 'Select a User' and shows a list of users, with the first entry being '1: admin administrator - admin@gmail.com'. To the right, there is a 'Create a User' form with the following fields:

- First Name: Sahil
- Last Name: Sandhu
- Phone Number: 7778889999
- Email: sahil@gmail.com
- Address: 1 test street
- City: Caledon
- Province: Ontario
- Password: sahil

An 'Add User' button is located at the bottom right of the form.

The admin panel has the ability to add new users, edit existing users, edit user bookings, add new cars, edit existing cars, and view business KPIs.

Car Rental Admin Panel

Edit Users Edit Cars View KPIs Name Here Exit to Dashboard Logout

Select a User

1: admin administrator - admin@gmail.com

7: Sahil Sandhu - sahil@gmail.com

Editing admin administrator

First Name	admin	Address	admin road
Last Name	administrator	City	Caledon
Phone Number	1112223333	Province	Ontario
Email	admin@gmail.com	View User Bookings	

[Delete](#)[Save](#)

Car Rental Admin Panel

Edit Users Edit Cars View KPIs Name Here Exit to Dashboard Logout

Sahil Sandhu's Bookings

1: adm

7: Sah

Create a Booking

Select a Car

2: 2003 Ferrari 360

4: 2003 Lamborghini Gallardo

3: 2017 Chevrolet Suburban

Total Price: -

Total Price Override

Start Date

End Date

[Add To User's Cart](#)[Add Booking](#)

Each user has their own booking menu where their bookings can be edited or new ones can be added directly to their account or to their cart.

Create a Booking

Select a Car

2: 2003 Ferrari 360
4: 2003 Lamborghini Gallardo
3: 2017 Chevrolet Suburban

Total Price: \$900

Total Price Override

900

Start Date

2025-12-03

End Date

2025-12-05

Price is automatically calculated depending on how many days the specific car is rented for. The admin can also override the price by editing the textbox.

Select a Car

2: 2003 Ferrari 360
4: 2003 Lamborghini Gallardo
3: 2017 Chevrolet Suburban

Total Price: \$900

Total Price Override

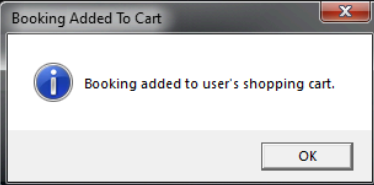
900

Start Date

2025-12-03

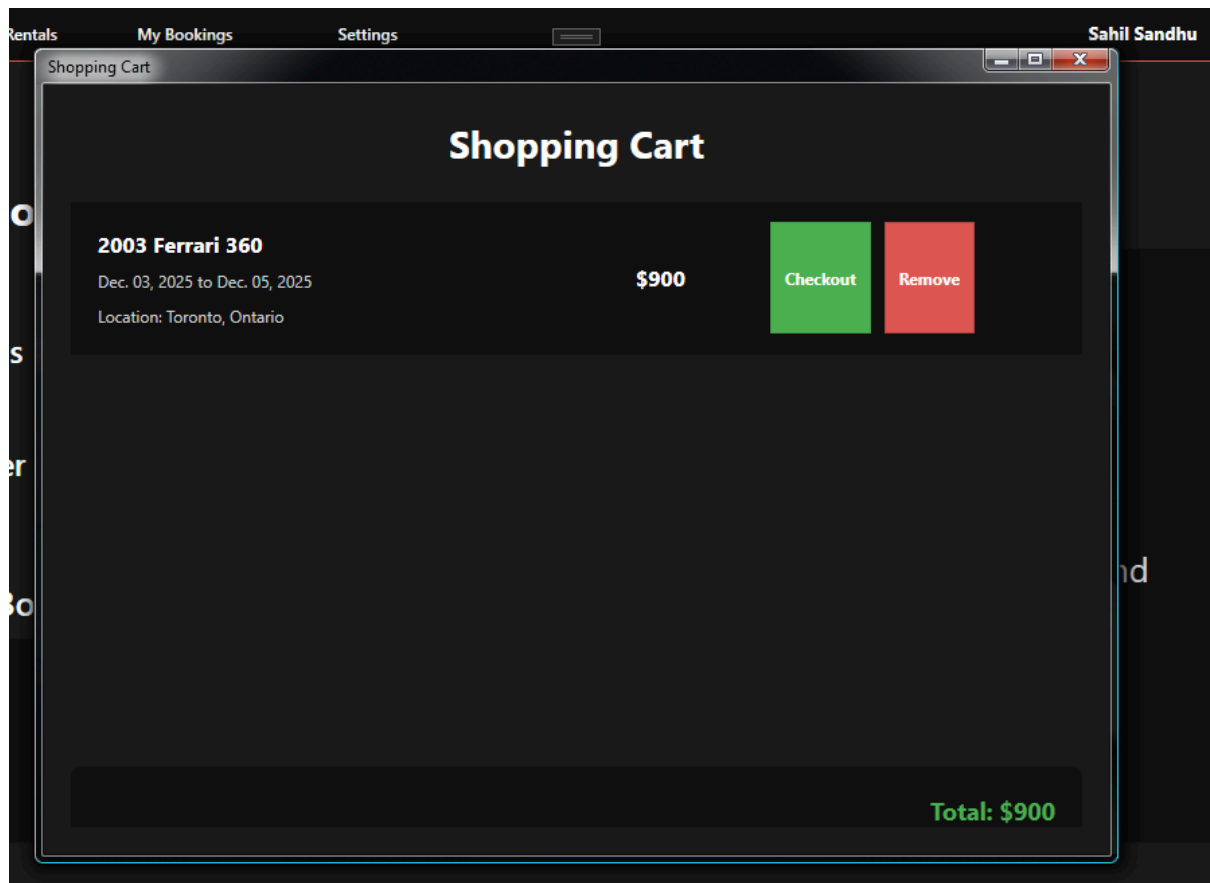
End Date

2025-12-05

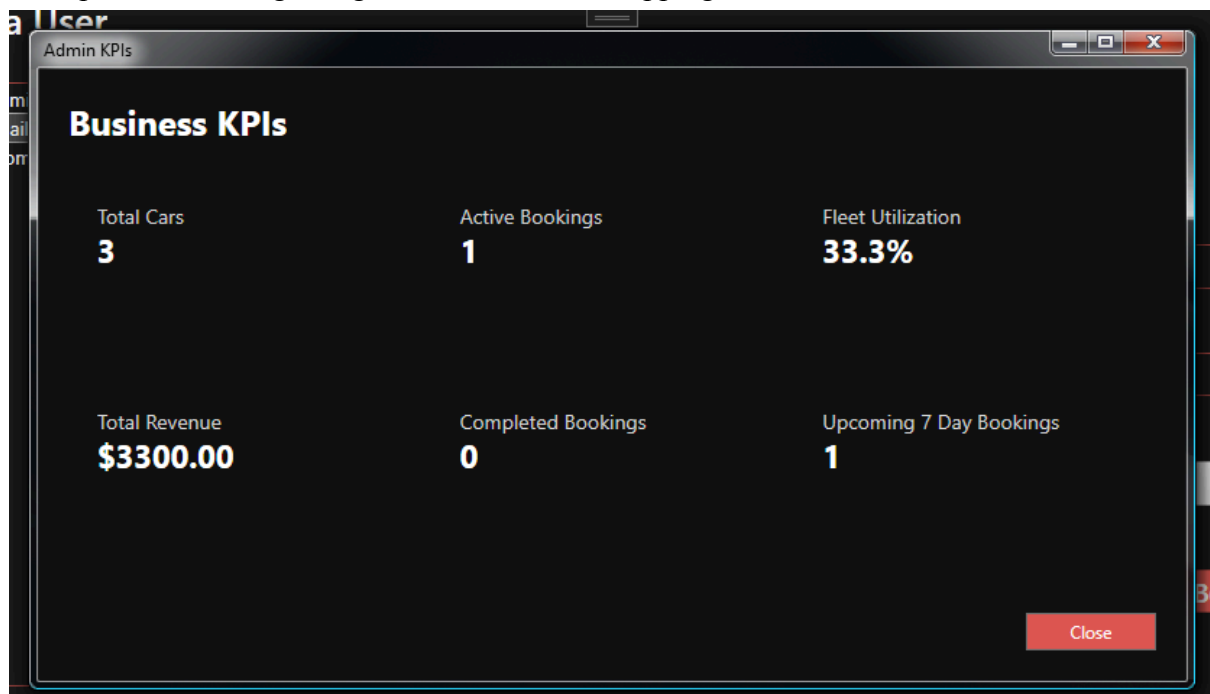


Add To User's
Cart

Add Booking



Example of a booking being added to a user's shopping cart.



The KPIs window displays various useful metrics.

The screenshot shows a web application titled "Car Rental Admin Panel". At the top, there are navigation links: "Edit Users", "Edit Cars", "View KPIs", "Name Here", "Exit to Dashboard", and "Logout". The main content area is divided into two sections. On the left, under the heading "Select a Car", there is a list of cars: "2: 2003 Ferrari 360", "3: 2017 Chevrolet Suburban", and "4: 2003 Lamborghini Gallardo". The third option is selected. On the right, under the heading "Editing 2003 Lamborghini Gallardo", there is a form with the following fields: "Make" (Lamborghini), "Price" (300), "Model" (Gallardo), "City" (Vancouver), "Year" (2003), "Province" (British Colombia), "Vehicle Type" (2-Door Coupe), and "ImageUrl (Web Image URL) (Optional)" (https://limitedslipblog.com/wp). There is a small image of an orange Lamborghini Gallardo next to the URL field. At the bottom of the form, there are two buttons: "Delete" (red) and "Save" (green).

The car editor is virtually the same as the user editor.

In this sprint, an admin panel and a KPI viewer were added. The admin panel tries to emulate all the features from an SQL editor that would be needed for this program. Features include Adding users, editing users, deleting users, viewing user bookings, editing user bookings, adding user bookings to either cart or account, deleting user bookings, adding cars, editing cars, and deleting cars. The admin panel can only be accessed by accounts flagged as admin accounts, which can only be done through an SQL editor for security. The other module added was the KPI viewer. The KPI viewer displays many useful statistics from data pulled directly from the SQL database.

Video Demonstration:

https://drive.google.com/file/d/1ezGKrXfbzFhIB7l_J_KQJM-4bVjOT0lq/view?usp=sharing

6. Testing and Review

Integration and System-Level Testing Summary

During Sprint 3, comprehensive integration and system-level testing was conducted to ensure all modules function cohesively. The primary focus was on the data flow between the new Admin Portal, the existing user-facing features, and the database. The results are summarized below:

Test Scenario	Modules Involved	Test Result	Notes
Admin Authentication & Access Control	Admin Login UI, User Database, Session Management	PASS	Admin accounts successfully access the portal; standard user accounts are correctly denied access.
Vehicle CRUD Operations	Admin Portal, Fleet Database, User Search UI	PASS	Vehicles added, edited, or deleted by an admin are immediately reflected in the user's search and vehicle details pages.
User Management via Admin	Admin Portal, User Database, User Dashboard	PASS	Admins can create user accounts and modify user data, which is correctly displayed in the respective user dashboards.
Booking & Cart Management	Admin Portal, Booking Logic, Shopping Cart, User Dashboard	PASS	Admins can create and edit bookings for users, which appear correctly in the user's cart or booking history.

Search Filter Integration	Search UI, Filter Logic, Backend API, Vehicle Database	PASS	All filters (category, price) correctly narrow down search results without breaking the existing date/location search.
End-to-End Booking Flow	User Login, Search, Cart, Payment, Booking Confirmation, KPI Update	PASS	A successful booking increments the reservation count and updates revenue/fleet utilization KPIs on the admin dashboard.
Simulated Payment Gateway	Payment UI, Booking Logic, Database	PASS	The payment form validates input, and upon "success," a booking is permanently created in the database. Failed payments prevent booking confirmation.
KPI Data Accuracy	Booking Database, KPI Calculation Logic, Admin Dashboard UI	PASS	Metrics like total bookings, revenue, and fleet utilization are accurately calculated and displayed in real-time.